

Phased dynamic analysis and prediction of rice *Chilo suppressalis* integrating remote sensing physiological indices and environmental factors

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Highlights

- Introduced LOESS smoothing and first-derivative analysis to identify the key developmental stages of *Chilo suppressalis*.
- Determined the screening scheme for sensitivity parameters specific to each pest development stage.
- Constructed a stage-specific daily dynamic prediction framework to achieve spatiotemporally explicit extrapolation of *C. suppressalis* population dynamics at the regional scale.

Abstract

Chilo suppressalis is a major pest in rice-producing regions, posing serious threats to rice yield and quality. Existing pest prediction research generally ignore the stage-specific heterogeneity in population dynamics and neglect the synergistic effects among meteorological, soil, and rice physiological information. This makes it challenging to accurately characterize the complete dynamic changes in pest populations. In this study, we explicitly highlight three methodological innovations: (1) the use of a LOESS-based curve fitting and slope-change detection framework to objectively partition *C. suppressalis* population dynamics into three stages—population establishment, expansion, and outbreak; and (2) the integration of multi-source data, including trap monitoring, rice physiological indices derived from remote sensing, and ERA5 meteorological and soil variables, to construct stage-specific prediction models.; and (3) building upon this stage-based framework, we designed a targeted sensitivity-parameter screening scheme and developed daily dynamic prediction models using the Random Forest (RF) algorithm, which incorporate meteorological, soil, and crop physiological indicators. The results demonstrate that the proposed stage-specific prediction model achieves excellent performance. During the outbreak stage, the R^2 for all three experimental fields exceeds 0.9, with MAE below 23.16 and RMSE under 32.86. In the Jiulong field, stage-specific predictions show R^2 values above 0.89. Compared with Long Short Term Memory (LSTM) and Prophet models, RF exhibits superior stability and generalization, with test set R^2 consistently above 0.69, highlighting its robustness and reliability for stage-specific prediction of *C. suppressalis* population dynamics. These findings highlight the practical value of our approach for enhancing comprehensive pest forecasting and supporting targeted pest management.

Keywords: rice, *Chilo suppressalis*, spatiotemporal prediction, random forest, stage-specific response

1. Introduction

Rice, as one of the most important staple crops globally, serves as the primary food source for over 3.5 billion people (Gülmez 2024; Mukherjee *et al.* 2025). The Food and Agriculture Organization of the United Nations (FAO) latest forecast predicts that global rice production is expected to reach 555.6 million tons (t) (FAO 2025), and its production stability is directly related to global food security and sustainable agricultural economies (Wang *et al.* 2022). However, affected by factors such as climate change and the intensification of cultivation practices, global crop losses due to pest damage reach up to 40% annually, with economic losses exceeding \$220 billion (Anonymous 2021), among which the yield gap caused by rice diseases and pests is particularly prominent. The National Agricultural Technology Extension and Service Center of China predicts that rice pest will generally occur severely in 2025, with an affected area of approximately 86.02 million hectares (ha), of which the affected area of rice *Chilo suppressalis* will be 16.67 million ha (Liu *et al.* 2025).

Chilo suppressalis was listed in China's "Category I Crop Diseases and Pests Directory" in 2020 and has the characteristic of secondary occurrence, capable of causing continuous damage to rice at different growth stages (Ge *et al.* 2025). During the tillering stage, it causes dead leaf seedlings. During the booting stage, it forms dead heart seedlings. During the heading stage, it causes white panicles and lodging. This is particularly problematic in rice-growing areas where pesticide resistance is increasingly severe, such as China (Mo *et al.* 2025), Japan (Ishiguro *et al.* 2007), and Iran (Majidi-Shilsar *et al.* 2023). *Chilo suppressalis* impedes water and nutrient transport by boring into rice stems and destroying vascular bundles, severely affecting normal rice growth, and can lead to rice yield reductions of 30 to 50% (Gao *et al.* 2020; Mao *et al.* 2022). In severely affected fields, total crop failure may even occur. Therefore, constructing accurate pest forecasting models not only provides scientific guidance to plant protection departments in formulating targeted control strategies, but also is a strategic requirement for achieving pest-resilient grain production and safeguarding national food security.

As a major stem-boring pest of rice, *C. suppressalis* requires accurate prediction of its occurrence dynamics as a key component for achieving integrated pest management in agriculture. Current research on crop pest forecasting and early warning systems primarily focuses on assessment of annual overall risk (Huang *et al.* 2019; Makori *et al.* 2024), and short-term predictions of one week (Ibrahim *et al.* 2022; Chacón-Maldonado *et al.* 2025) or one month (Ma *et al.* 2019; Skawsang *et al.* 2019). These approaches typically analyze and model the entire pest occurrence period as a single, unified object. However, *C. suppressalis* exhibits significant stage-specific characteristics during its actual occurrence process, with pest population showing multi-peak fluctuations in time series, where each peak corresponds to the generational succession process. Taking the Yangtze River Basin as an example, *C. suppressalis* typically completes three to four generations

per year (Chen H *et al.* 2020; Pan *et al.* 2025). The first generation usually occurs from early April to early June, representing the concentrated emergence period of larvae following the overwintering adult moths' reproduction. The second generation mainly appears from late June to late July, with significantly increased pest density, constituting an important stage for annual dispersal. The third and fourth generations predominantly occur from August to October, typically forming the peak stage of annual pest occurrence. By late October, mature larvae gradually migrate to overwintering sites such as rice stubble and enter diapause to survive the winter (Huang *et al.* 2020; Cao *et al.* 2025). Therefore, treating the entire occurrence process as a homogeneous stage for unified modeling cannot effectively capture the inherent stage-specific variation patterns of *C. suppressalis*, thereby limiting the timeliness and accuracy of forecasting results. To address this issue, this study introduces a LOESS-based curve fitting and slope variation analysis framework to divide the occurrence process of *C. suppressalis* into distinct developmental stages based on pest monitoring data and crop phenological rhythms, and constructs stage-specific dynamic prediction models, aiming to enhance the prediction accuracy of pest dynamics at each critical stage and provide scientific support for region-scale pest monitoring and precision management.

Crop pest outbreaks are usually closely related to soil environment, meteorological conditions, and crop physiological status. Among these, the soil environment plays a fundamental role in regulating pest population dynamics, particularly through soil temperature and soil moisture, which affect pest survival, development rate, diapause termination, and overwintering success (Poggi *et al.* 2018; Wu *et al.* 2021). Meteorological factors such as air temperature and humidity further modulate pest activity and dispersal, and meteorology-driven forecasting models have therefore become essential tools for agricultural pests control (Caubel *et al.* 2017; Sriwanna 2022; Xian *et al.* 2023). Meanwhile, crop physiological indicators that reflect canopy growth status and stress responses have been widely applied to identify potential high-risk areas (Skawsang *et al.* 2019; Ruan *et al.* 2022; Allu and Mesapam 2025; Li *et al.* 2025), providing key biological information for pest prediction. Despite their critical roles, soil environmental factors remain underrepresented in current pest prediction studies. Existing research on sensitive parameter selection primarily focuses on meteorological factors and crop physiological indicators, while rarely incorporating soil environmental factors into a unified analytical framework. Moreover, these studies lack systematic exploration and joint modeling of the synergistic interactions among meteorological, soil, and crop physiological factors. In addition, most studies emphasize the overall effects of sensitive parameters on pest occurrence, neglecting the dynamic changes of dominant factors across different developmental stages. This limitation makes it difficult to reveal the stage-specific dynamics of driving factors during pest occurrence. In view of this, based on the stage-specific features of the pest occurrence process and from the perspective of synergistic effects of multi-source environmental factors, this study systematically analyzes the dominant effects at different stages, which is significant for improving the timeliness, explanatory power, and scientific

decision-making support of prediction models.

Crop pest prediction models are playing an important role in assisting plant protection experts in assessing pest epidemic trends and implementing coordinated control strategies. Traditional forecasting methods often rely on expert knowledge and statistical techniques (Huang *et al.* 2025), which are time-consuming and labor-intensive (Domingues *et al.* 2022; Gao *et al.* 2025). In addition, their predictive performance is frequently constrained by fixed empirical rules and linear assumptions, making it difficult to capture the complex, nonlinear relationships between forecasting variables and pest dynamics (Tan *et al.* 2021). In contrast, machine learning models, with their powerful data integration capabilities, data processing, and predictive accuracy, are more effective in revealing the complex relationships between pest occurrence and multi-source environmental variables (Durai and Shamili 2022; Waters *et al.* 2025), providing timely and accurate pest early warnings for agricultural production and demonstrating great potential in crop pest forecasting. However, current crop pest prediction and early warning primarily focus on determining whether pests occur (He *et al.* 2024), occurrence probability (Uusitalo *et al.* 2019; Neta *et al.* 2023), or severity level (Jiang *et al.* 2021; Chen *et al.* 2022). These studies primarily rely on observing crop health conditions to indirectly infer the pest occurrence (Ye *et al.* 2025). For example, the severity of infestation is often classified into four levels (none, low, moderate, and severe) based on visible crop damage (Chen P *et al.* 2020). There is relatively insufficient continuous and detailed analysis directly targeting pest population dynamics, and dynamic forecasting frameworks specifically tailored to the occurrence characteristics of *C. suppressalis* are lacking. As a result, it is difficult to comprehensively reveal the temporal variation patterns of its occurrence dynamics. To address this gap, we construct stage-specific dynamic prediction models for the daily abundance of *C. suppressalis* using the random forest (RF) algorithm, which integrate multi-source meteorological, soil, and crop physiological information. Thereby enhancing the prediction capability of dynamic processes and providing a technical pathway for achieving more precise and efficient pest control.

In summary, existing studies have neglected the stage-specific heterogeneity in pest occurrence dynamics, with prediction factor selection mostly based on static correlations throughout the entire pest occurrence cycle, while lacking dynamic forecasting mechanisms coupled with crop phenological rhythms. These limitations hinder the precise tracking of the full temporal succession of pest populations. Therefore, the main objectives of this study include: (1) identifying the key developmental stages of *C. suppressalis* by characterizing pest population dynamics using a LOESS-based curve fitting and slope-change analysis framework; (2) systematically screen stage-specific sensitive environmental parameters, with explicit consideration of soil, meteorological, and crop physiological factors; and (3) constructing a daily-scale, stage-specific dynamic prediction model based on the random forest algorithm to enable continuous forecasting of pest population dynamics and comprehensively characterize the occurrence patterns of *C.*

suppressalis.

2. Materials and methods

2.1 Experimental site

As a key rice-producing region on the southern flank of the Yangtze River Delta in China, Ningbo City features a composite topography combining plain river networks (40.3%) and low hills and mountainous terrain (59.7%) (Fig. 1-B). Under the influence of a subtropical monsoon humid climate, it forms a unique rice-growing habitat with an average annual temperature of 17.5°C, combined with 1,716 h of annual sunshine and a frost-free period of 250–270 days. Together, these climatic conditions collectively provide abundant light and heat resources for rice cultivation (Zhou *et al.* 2022). This endows it with high yield potential (with an average yield of 7.73 t/ha), but also brings ecological risks. The annual precipitation of 1,539 mm with over 60% concentrated between May and September coincides with key stages of the rice growth period, creating high temperature and humid conditions conducive to pest outbreaks (Zhai *et al.* 2024).

Ningbo City has diverse rice cultivation patterns, with four major types coexisting in an interlaced manner: single-crop early rice (early April to late July), single-crop middle rice (early May to late September), single-crop late rice (mid-June to late October), and double-crop rice (early April to late October) (Fig. 1-C), constituting a spatiotemporally continuous host supply system. This artificially created continuous food supply condition, combined with the high resistance of *C. suppressalis* to multiple pesticides (Lu *et al.* 2017), jointly promotes a high risk situation where *C. suppressalis* exhibits overlapping generations and continuous occurrence.

Given the representative rice cultivation patterns and the outbreak characteristics of *C. suppressalis*, this study selected Ningbo City as the experimental area. Continuous field monitoring was established at three sites: Jiulong field, Xiandai field, and Mailang field (Fig. 1-B). Each site has approximately 30.0 ha of rice cultivation, forming a landscape-continuous sampling framework. Among these, Jiulong field adopts a single-crop middle rice system, while Yinzhou and Fenghua follow a double-crop rice system, representing the mainstream rice cultivation systems in the region.

2.2 Data acquisition and processing

Spatial distribution data of rice cultivation Based on a total of 120 Sentinel-1A Interferometric Wide (IW) mode SAR high-resolution (10m) Level-1 GRD (Ground Range Detected) data products covering Ningbo City in 2024, a rice identification algorithm developed by our team was applied, which integrates SAR backscattering features, topographic features, statistical characteristics, and phenological parameters.

This algorithm enabled accurate identification of rice cultivation areas in Ningbo City (Fig. 1-C), with an overall accuracy of 0.91 (Wang *et al.* 2024). The method fully leverages the advantages of multi-temporal SAR remote sensing in capturing the dynamic responses of rice throughout the entire growth cycle, and is well-suited to the region's complex cropping patterns. It provides highly reliable spatiotemporal rice distribution data to support subsequent *C. suppressalis* forecasting models.

Field monitoring data The developmental process of *C. suppressalis* consists of four stages: egg, larva, pupa, and adult (Fig. 2-A). In three experimental fields located in Jiulong, Xiandai, and Mailang, TPXY-SAM4.0 IoT-based pheromone trapping and pest monitoring system were deployed to monitor the occurrence dynamics of *C. suppressalis* adults (Fig. 2-B). These systems integrate intelligent pheromone trap terminals with multi-parameter meteorological sensors, enabling continuous in-situ monitoring of *C. suppressalis* population dynamics and associated Environmental factors in rice fields from 2023 to 2024. The multi-parameter meteorological sensors monitor real-time atmospheric and soil data, with data collected at hourly and daily resolutions. The collected air temperature and *C. suppressalis* population dynamics data from April to October 2024 are shown in Fig. 2-C. The population dynamics of *C. suppressalis* can be clearly characterized by four typical developmental stages: population establishment, expansion, outbreak peak, and overwintering preparation.

In addition, the pheromone trap devices are equipped with a built-in high-definition camera that can automatically capture pest images at scheduled each day, as well as support remote manual control for photography. In this study, automatic image acquisition was scheduled at 10:30 AM every day to identify and count *C. suppressalis* (Fig. 3), with an image recognition accuracy of 97.2% (Shen *et al.* 2025).

ERA5 meteorological and soil factors Daily Environmental variables for Ningbo City during the periods April 1 to October 31 in 2023 and 2024 were obtained from the ERA5 reanalysis dataset. The extracted variables include both meteorological and soil factors. Meteorological factors include 2 m air temperature (AT), relative humidity (RH), 2 m dewpoint temperature (DT), precipitation (P), wind speed (WS), wind direction (WD), and atmospheric pressure (AP). Soil factors include topsoil (0-7 cm) temperature (ST) and volumetric soil moisture (SM). All data were stored as 32-bit floating point and referenced to the Web Mercator projection. The original grid resolution of $0.1^{\circ} \times 0.1^{\circ}$ was resampled to 500 m resolution using bilinear interpolation (eq. (1)), resulting in a spatiotemporally continuous multi-dimensional data cube (Fig. 4). This procedure avoids scale-related inconsistencies during model training and ensures spatial coherence of the predictions.

$$f(x, y) = \frac{(x_2 - x)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)} f(x_1, y_1) + \frac{(x - x_1)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)} f(x_2, y_1) \\ + \frac{(x_2 - x)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)} f(x_1, y_2) + \frac{(x - x_1)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)} f(x_2, y_2) \quad (1)$$

where (x, y) denotes the target location at which the function value is to be estimated. $f(x_1, y_1)$, $f(x_2, y_1)$, $f(x_1, y_2)$, $f(x_2, y_2)$ are the known function values at four neighboring points, (x_1, y_1) , (x_2, y_1) , (x_1, y_2) , (x_2, y_2) are the coordinates of these known points.

Rice remote sensing physiological indicators Rice remote sensing physiological indices, serving as quantitative indicators reflecting crop physiological status, can effectively monitor and assess the growth condition of the rice canopy and its response to pest stress. Based on the MODIS Terra MOD09GA surface reflectance product, this study obtained daily-scale data at 500 m spatial resolution for the rice growing seasons of 2023 and 2024. Cloud masking was applied to remove pixels affected by cloud interference, and forward-filling methods were used to repair missing data to ensure the continuity of the time series data. The Savitzky-Golay filtering method was applied with a window size of 7 and polynomial order of 3 to smooth 11 rice remote sensing physiological indices and environmental indices (Table 1), reducing noise interference and optimizing the vegetation temporal profiles throughout the rice growth cycle. This approach extracted stable and reliable temporal features representing rice physiological status, comprehensively reflecting canopy structure, chlorophyll content, plant senescence, as well as water and soil background characteristics. These features characterize the physiological and ecological traits of rice at different growth stages and its growing environment, providing high-quality remote sensing physiological data inputs for subsequent *C. suppressalis* occurrence prediction models.

2.3 Research framework

To address the precise prediction needs for spatiotemporal dynamics of *C. suppressalis* population in regions with complex rice planting structures, this study constructed stage-specific dynamic prediction models based on sex pheromone trapping data, rice remote sensing physiological indices, ERA5 meteorological and soil data (Fig. 5). The method mainly includes the following four aspects: (1) Based on rice phenological periods and the pest's ecological rhythms, locally estimated scatterplot smoothing (LOESS) curve fitting and slope change analysis were introduced to delimit the key stages of the *C. suppressalis* population dynamics; (2) Combining Pearson correlation coefficients with the Boruta algorithm, sensitivity parameters for different stages of *C. suppressalis* were determined, and the SHapley Additive exPlanations (SHAP) method was further utilized to explain the relative contribution of each factor to *C. suppressalis*; (3) Integrating meteorological, soil, and crop physiological factors, prediction models for different population stages of *C. suppressalis* were constructed, generating the pest spatiotemporal distribution prediction maps and (4) the prediction accuracy of the models were validated and evaluated.

2.4 Identification of *C. suppressalis* population dynamics stages

The population dynamics of *C. suppressalis* exhibit a multi-peak variation pattern of "gradual-expansion-outbreak-stabilization" throughout the entire rice growth period. This characteristic is mathematically manifested as stage-specific fluctuations in the slope of the population curve. These slope variations enable segmented identification through trend analysis of fitted curves, which provides a theoretical basis for the quantitative division of pest occurrence stages. To clarify the specific occurrence stages of *C. suppressalis* within the rice growth cycle, this study comprehensively considered variations in rice cropping systems, rice phenological stages, and the pest's succession patterns, combined with field pest monitoring data, to conduct stage division of the occurrence process.

To address the nonlinearity, high volatility, and multi-peak characteristics inherent in the occurrence process of *C. suppressalis*, this study employed a smoothing method based on LOESS. The method was applied to fit the trend of the *C. suppressalis* population over time in three experimental fields. This approach effectively filtered out high-frequency noise while preserving key trends in population dynamics. The smoothing parameter frac was set to 0.1 to balance the smoothness of the curve with the retention of local details.

Building upon the fitted curves, the first derivative (i.e., slope) was calculated to characterize the dynamic rate of change in the *C. suppressalis* population. Given the significant differences in population expansion rates across developmental stages, slope variation served as a sensitive indicator for identifying critical inflection points. LOESS smoothing was applied to extract the underlying population trend while filtering short-term monitoring noise. An excessively low polynomial order (e.g., 2 or 3) tends to oversimplify the data and fails to adequately capture key turning points such as rapid population growth or decline. Conversely, too high an order increases the risk of overfitting random fluctuations and introducing spurious noise. Therefore, a fifth-order polynomial was adopted in this study to balance flexibility and robustness. This ensures the model can effectively delineate nonlinear growth patterns over the pest life cycle while filtering short-term monitoring noise, thereby accurately extracting biologically meaningful, stage-related population trends. Subsequently, a local extrema detection method with order of 5 was applied to the slope sequences, from which the two local minima with the most significant amplitude changes were selected as turning points in the pest occurrence process.

Finally, based on the two identified turning points, the occurrence process of *C. suppressalis* during the rice growing season was divided into three distinct developmental stages: the establishment stage, the expansion stage, and the outbreak stage.

2.5 Determination of the sensitive parameters

The occurrence of *C. suppressalis* is jointly driven by multiple environmental factors, including meteorological, soil, and crop physiological indicators. However, these factors often exhibit strong redundancy and multicollinearity, which not only obscure the effects of key driving variables, but also reduce the computational efficiency and predictive accuracy of the model.

Based on the typical population succession process of *C. suppressalis*, comprising population establishment, expansion, and outbreak stages, this study adopted a stage-specific approach to determine sensitive parameters, aiming to enhance the biological relevance and statistical robustness of variable selection. For each stage-specific dataset, Pearson correlation coefficients were calculated, high inter-correlation ($|r| > 0.8$) were eliminated to preliminarily reduce multicollinearity. Furthermore, the Boruta algorithm was applied to evaluate the importance of the factors retained after correlation-based screening. By systematically comparing the importance of original factors with that of randomly generated shadow factors, this algorithm identifies key factors that significantly contribute to the population dynamics of *C. suppressalis* at different stages. To enhance the spatiotemporal generalizability of the sensitive parameters, those identified as important by Boruta in at least two stages were selected to construct a unified set of sensitive parameters, ensuring strong driving capacity across multiple developmental stages.

To gain deeper insight into the role of these cross-stage robust sensitive parameters in model prediction, this study employed the SHAP method to quantify factor importance and interpret their contributions within each stage-specific dataset. This approach provides interpretability support for the pest forecasting models by revealing the underlying mechanisms through which the selected factors influence model outcomes.

2.6 Construction of the pest forecasting model

To achieve spatiotemporal prediction of *C. suppressalis* population, this study integrated meteorological and soil factors, rice remote sensing physiological indices, and in situ field monitoring data. RF offers significant advantages in ecosystem modeling by effectively capturing nonlinear relationships, handling high-dimensional feature spaces, and demonstrating robustness against outliers and multicollinearity. Considering that this study involves heterogeneous multi-source data, complex interactions among variables, and nonlinear responses of pest populations to environmental factors, RF was selected as the modeling approach. It was used to develop stage-specific dynamic prediction models for *C. suppressalis*, combining biological characteristics of pest developmental stages with RF regression to achieve accurate and reliable population forecasts.

Based on the stage-specific division of *C. suppressalis* into population establishment, expansion, and outbreak stage, the data for each stage were split into training and testing sets at an 8:2 ratio to ensure independence between model training and performance evaluation. Additionally, the datasets with sensitive

parameters selected in Section 2.5 were used as model input variables, while the observed abundance of *C. suppressalis* from pest monitoring sites served as the target variable.

Model training was conducted on Python 3.12, with the RF regression model. Four key hyperparameters of the RF model were optimized through a grid search strategy. The hyperparameters search ranges are listed in Table 2. A three-fold cross-validation strategy was employed, with the optimization objective of minimizing the negative root mean squared error (RMSE), to determine the optimal hyperparameter combinations for each developmental stage of *C. suppressalis* population dynamics. The stage-specific daily datasets of sensitive parameters determined in Section 2.5 were input into the trained prediction models to generate spatial distribution maps of *C. suppressalis* population dynamics.

2.7 Assessment of prediction model accuracy

Model performance evaluation metrics include the Determination Coefficient (R^2), RMSE, and Mean Absolute Error (MAE) to comprehensively assess the predictive accuracy and robustness of the model across different stages and regions. The calculation formulas for each evaluation metric are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (3)$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (4)$$

where n is the number of samples, y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ represents the mean of the observed values.

3. Results

3.1 *Chilo suppressalis* population stage

The establishment stage represents the initial establishment of the *C. suppressalis* population driven by post-diapause emergence and low-density larval accumulation. *C. suppressalis* overwinters as mature larvae within rice stubble. In the following spring, when temperatures stabilize above 14°C (typically from early to mid-April), the overwintering larvae gradually terminate diapause, pupate, and emerge as adults. In Ningbo, the first-generation adults of *C. suppressalis* begin sporadic oviposition during the tillering stage of early rice

(late April to early May). Approximately 7 to 10 days later, the eggs hatch, and the larvae begin boring into rice stems, gradually causing visible symptoms such as dead leaf sheaths in the field. At this stage, the rice plants are still in the early phase, with relatively low planting density. Consequently, oviposition is scattered and pest sources are widely distributed, resulting in an overall low larval density and slow population growth. Thus, the *C. suppressalis* population is in a stage of gradual baseline accumulation. Based on the LOESS-smoothed curves of cumulative *C. suppressalis* abundance and their first-order derivatives, the first significant inflection was identified on June 14 in Jiulong field (Fig. 6-A and B), June 13 in Xiandai field, and June 11 in Mailang field (Fig. 6-C–F). Therefore, the establishment stage of the *C. suppressalis* population during the rice growing season in Ningbo can be defined as the period from early April to early June.

The expansion stage is characterized by rapid population growth driven by favorable climatic conditions and increased host plant suitability. With the accumulation of the first-generation pest population, *C. suppressalis* entered the expansion stage, during which a marked increase in population was observed. In Ningbo, the onset of the plum rain season typically occurs in early June. The high humidity during this period can suppress the development of *C. suppressalis* to some extent (Ding *et al.* 2019). However, after the end of the plum rain season in early July, the weather turns hot, dry, and sunny—conditions that significantly favor adult activity, mating, and oviposition. Meanwhile, rice fields are in a transitional phase between single and double cropping systems, with most plants in the tillering to panicle initiation stages. During this period, the rice leaf sheaths become softer and looser, with expanded intercellular spaces, providing a favorable oviposition substrate and entry pathways for larval invasion. These conditions collectively accelerate the accumulation and expansion of *C. suppressalis* population density. As shown in Fig. 6, the end of the population expansion stage occurred on Aug. 2 in Jiulong field and Aug. 4 in Xiandai field, while the turning point appeared earlier on July 24 in Mailang field. Therefore, the population expansion stage of *C. suppressalis* during the rice growing season in Ningbo can be defined as the period from early June to the end of July.

The outbreak stage represents the annual population peak of *C. suppressalis*, characterized by explosive growth and severe crop damage. As middle rice enters the heading to grain-filling stage and late rice begins the booting stage, *C. suppressalis* population undergo rapid growth and concentrated outbreak. During this period, the leaf sheath tissues of rice plants become soft and loose, providing ideal habitats and oviposition sites for *C. suppressalis* adults, accelerating the accumulation and reproduction of the pest population. Large numbers of adults are highly active in the field with intensive egg-laying, and the hatched larvae quickly bore into rice stems, causing typical damage symptoms such as deadheart, dead leaf sheaths, and white heads. This leads to a sharp increase in field population density and significantly intensified pest damage, with *C.*

suppressalis population reaching their annual peak. Therefore, the outbreak stage of *C. suppressalis* during the rice growing season in Ningbo is defined as the period from early August to late October.

The overwintering preparation stage represents the endpoint of the annual pest occurrence process of *C. suppressalis*, driven by host depletion and unfavorable climatic conditions. As middle rice and late rice are successively harvested in early and late October, respectively, rice plants disappear in large quantities, host resources rapidly decrease, and with continuously declining temperature, the development activity of *C. suppressalis* are suppressed, hindering further expansion of the pest population. During this stage, the rate of change in pest population shows an initial decline followed by stabilization, population quantity growth essentially stagnates, and the population overall enters a decline period, gradually transitions into overwintering state. Considering that pest dynamics during this stage tend to stabilize with no significant dynamic fluctuations, this study no longer conducts further modeling or driving factors analysis for this period, but merely identifies it as the endpoint of the annual pest occurrence process.

3.2 Sensitive parameters

This study employed a dual screening strategy combining correlation analysis and the Boruta algorithm to identify 11 key predictive variables, including four meteorological-soil factors and seven rice remote sensing physiological indices. The analysis revealed that EVI, SIPI, and NDWI serve as core predictive variables throughout the entire population cycle, while BSI, AP, SM, and the combination of AP and WS emerge as dominant factors during the expansion and outbreak stages, respectively. The stage-specific analysis of driving mechanisms indicates that rice canopy structure, water status, and specific environmental conditions synergistically regulate pest population dynamics.

The results of sensitive parameter selection using a dual statistical screening approach are presented in Table 3. AT, GNDVI, NDVI, and SAVI were excluded due to Pearson correlation coefficients greater than 0.8 with other factors across all three stages (Fig. 7), indicating strong multicollinearity among these variables. Subsequently, the Boruta method further eliminated factors with no significant contribution to the model. Among these, WD consistently showed low importance in all three stages and was unable to effectively explain the variation in *C. suppressalis* population. EVI, SIPI, and NDWI were retained as parameters throughout all three stages. The final sensitive parameters dataset comprised four meteorological and soil factors (SM, DT, AP, and WS) and seven rice remote sensing physiological indices (RDVI, EVI, SIPI, BSI, PSRI, SR, and NDWI).

Furthermore, the SHAP method was employed to quantify the importance of sensitive parameters at different stages (Fig. 8). The results indicate that the dominant factors for the spatial distribution of *C. suppressalis* population differ significantly across different stages. During the initial population establishment stage, EVI

effectively reflected canopy closure and crop vigor in paddy fields, creating a favorable microhabitat for pest development with SHAP values significantly higher than other factors (Fig. 8-B).

Upon entering the population expansion stage, the importance of BSI, AP, and SM increased significantly (Fig. 8-C). Specifically, BSI reflects the degree of surface exposure, which is related to adult habitat and aggregation potential; lower AP promotes adult migration, oviposition, and source accumulation (Miao *et al.* 2021). Meanwhile, SM positively influences egg hatching and larval survival (Rhodes *et al.* 2022).

During the outbreak stage, the dominant role of AP was further amplified (Fig. 8-D), with the coupled effect of AP and WS impacting pest dispersal pathways and outbreak magnitude (Wu *et al.* 2019). Meanwhile, the SHAP values of RDVI, NDWI, and PSRI increased notably, indicating that rice vegetation structure and water status play important roles in regulating the spatial distribution patterns during the outbreak peak period.

Overall, the evolutionary patterns of stage-specific dominant factors reveal the synergistic driving mechanism between rice remote sensing physiological indices and environmental factors in shaping the population dynamics of *C. suppressalis*. This provides a theoretical foundation and decision-making support for stage-specific dynamic forecasting and precision management of pest outbreaks.

3.3 Model performance

Stage-specific hyperparameter optimization ensured that the models were appropriately adapted to the varying complexity of *C. suppressalis* population dynamics across different stages. Through grid search and three-fold cross-validation, the optimal hyperparameter combinations for model at different stages were determined (Table 4). During the expansion stage, max_depth was set to 10 and n_estimators to 50. In both the establishment and outbreak stages, max_depth was set to None, giving average depths of 14.88 and 16.47, respectively, and n_estimators set to 200. For all three stages, min_samples_leaf and min_samples_split were fixed at 1 and 2, respectively.

Overall, the stage-specific models exhibited robust predictive performance and satisfactory generalization ability across both training and testing datasets. The model demonstrated strong fitting performance on the training sets across all three stages (Fig. 9-A), with R^2 values exceeding 0.96, indicating that the model effectively captured the relationship between features and pest population in the training data. Notably, the training errors in the establishment and outbreak stages were relatively low, with MAE values of 4.95 and 10.88 respectively, reflecting better fitting quality. On the testing sets (Fig. 9-B), the models for the establishment and the outbreak stages achieved R^2 values of 0.74 and 0.81, maintaining high explanatory power, whereas the expansion stage showed a slightly lower test R^2 of 0.67. Overall, these results validate the

effectiveness of the stage-specific variable selection and hyperparameter optimization strategies in enhancing the predictive stability and generalization ability of the model.

This study compared the performance of the Long Short-Term Memory network (LSTM), Prophet, and RF in predicting the population abundance of *C. suppressalis* across different stages (Fig. 10). As a representative recurrent neural network, LSTM is capable of capturing temporal dependencies and nonlinear dynamics in time series data through a sliding window structure. In this study, LSTM exhibited a certain fitting ability on the training set. For example, during the establishment stage, the training R^2 reached 0.74, indicating that the model could learn part of the temporal patterns. However, its predictive performance on the test set declined markedly, particularly during the outbreak stage, where the test MAE and RMSE increased substantially to 136.48 and 146.02, respectively. This result suggests that LSTM is sensitive to sample size and noise, and is prone to overfitting or limited generalization under small-sample and highly variable data conditions (Baste *et al.* 2025).

The Prophet model, as an additive time series forecasting approach, models temporal dynamics by decomposing the series into trend and seasonal components and exhibits relatively stable predictive performance across stages. The RMSE values for the three stages are 16.07, 29.36, and 54.82, respectively, indicating a certain degree of fitting stability in capturing stage-specific variations in *C. suppressalis* population abundance. Notably, during the establishment stage, Prophet achieved a test R^2 of 0.81, which is slightly higher than that of the RF model (0.76), suggesting an advantage in capturing relatively smooth population growth trends at this stage. However, as population dynamics become increasingly complex during stage progression, the predictive accuracy of Prophet gradually declines. During the expansion stage, the test R^2 decreased to 0.64, slightly lower than that of RF (0.69), and further dropped to 0.49 in the outbreak stage. These results indicate that Prophet still has limitations in representing complex nonlinear population dynamics driven by multidimensional environmental variables (Milli 2025).

In contrast, the RF model demonstrated consistently superior and more stable performance across all three stages. During the establishment stage, RF maintained good generalization ability on the test set (MAE=13.36, RMSE=17.88, R^2 =0.76). During the expansion stage and the outbreak stage, despite pronounced increases in population variability, RF still achieved relatively low prediction errors and high explanatory power, with test R^2 values of 0.69 and 0.72, respectively. These results highlight the robustness of RF under high-dimensional and nonlinear environmental conditions.

Overall, LSTM and Prophet show potential for time series modeling, but their predictive performance may be limited by small sample sizes and may require further optimization of model structure. In contrast, RF exhibited clear advantages under small-sample, high-dimensional, and nonlinear conditions, making it

particularly suitable for regional-scale, multi-stage prediction of *C. suppressalis* population abundance within the framework of this study.

3.4 Spatiotemporal characteristics of *C. suppressalis*

The predicted spatial distribution of *C. suppressalis* abundance during the 2024 rice growing period is shown in Fig. 11. Overall, the pest distribution exhibited pronounced temporal variations and spatial clustering, with distribution patterns highly correlated with rice cultivation conditions at different stages.

The spatial distribution of *C. suppressalis* in Ningbo exhibited a clear stage-dependent expansion pattern, transitioning from localized accumulation to rapid regional diffusion. The establishment stage of *C. suppressalis* spanned from April 4 to June 10. During this period, the population remained in the early accumulation phase, characterized by low overall abundance and a scattered distribution in early rice growing areas (Fig. 11-A). As early rice grew and middle rice was transplanted, the population gradually expanded into surrounding regions and exhibited a slow upward trend in abundance (Fig. 11-B and C). From June 12 to July 31, as rice gradually entered the tillering to panicle initiation stages, *C. suppressalis* simultaneously entered its population expansion stage. After mid-to-late July, as the climate shifted from the plum rain season to hot and dry conditions, the rate of pest expansion accelerated significantly. The abundance increased markedly, accompanied by a northward shift in spatial distribution and a multi-centered diffusion pattern (Fig. 11-E and F).

From August 15 to October 22, *C. suppressalis* entered its outbreak peak stage. During this period, both middle rice and late rice were generally in the critical reproductive stages, from panicle initiation to grain filling. The increased canopy closure and the soft, loose structure of the leaf sheaths created favorable conditions for adult habitat and oviposition, so the pest abundance rapidly accumulated, density rose sharply, and the abundance reached its annual peak (Fig. 11-G–I). With the successive harvest of middle rice and the gradual maturation of late rice, the paddy field ecosystem began to change, leading to a stabilization in both pest abundance and its spatial spread (Fig. 11-I). The dynamic spatiotemporal distribution characteristics of *C. suppressalis* highlight the significant role of rice cropping systems and planting structures on their succession patterns, providing a scientific basis for formulating stage-specific and region-targeted pest management strategies.

4. Discussion

4.1 Performance and advantages of prediction model

This study innovatively develops a multi-factor coupled prediction framework. By introducing the LOESS method for phenological stage division of *C. suppressalis* population dynamics and integrating multi-source

data—including rice remote sensing physiological indices, meteorological, and soil variables—the model effectively identifies the dominant driving mechanisms across different ecological stages, enabling high-precision spatiotemporal dynamic prediction. This framework not only provides reliable early warning information for regional pest occurrence trends but also offers actionable decision support for formulating stage-specific precision control strategies. A comparison between the *C. suppressalis* spatial prediction results and the actual pest monitoring data shows that the overall predicted trends have a high degree of agreement with the measured data, effectively capturing the process of pest population changes at various stages (Fig. 12). This demonstrates that the multi-factor coupled prediction model, constructed based on rice remote sensing physiological indices, meteorological and soil environmental factors, delivers strong spatial continuity and timeliness at the regional scale. It can effectively capture the accumulation, dispersal, and outbreak processes of *C. suppressalis* pest sources, supporting the dynamic identification of high-risk areas and the formulation of stage-specific and region-specific control strategies.

The *C. suppressalis* prediction model can accurately depict the dynamic dispersal process of pest sources across the population establishment, expansion, and outbreak stages in all three experimental fields (Fig. 13). During the population establishment stage, the model exhibited strong fitting performance in the Jiulong field and Mailang field, with R^2 values of 0.90 and 0.85, respectively (Fig. 13-A and G), indicating that the model possesses strong responsiveness during the initial pest source stage. In contrast, the predictive accuracy for the Xiandai field during this stage was relatively lower, with an R^2 of 0.44 (Fig. 13-D). This error may be attributed to the smaller pest population base, complex field landscape, and strong environmental heterogeneity, all of which hinder the model's ability to extract stable predictive factors.

During the population expansion stage, both the Jiulong field and Xiandai field maintained relatively high prediction accuracy, with R^2 values of 0.89 and 0.75, respectively (Fig. 13-B and E). Notably, the Jiulong field demonstrated stable performance, with an MAE of 10.24 and an RMSE of 17.05, indicating that the model possesses strong adaptability during the rapid accumulation stage of the pest population. The predictive performance for the Mailang field was comparatively weaker in this stage, with an R^2 of 0.47 (Fig. 13-H), and errors were concentrated in some sample points. Although the overall trend was consistent with observed data (Fig. 12), the limited sample size may have affected model stability.

During the population outbreak stage, the model exhibited significantly improved predictive performance, achieving high predictive accuracy across all three experimental fields. The R^2 values for the Jiulong field, Xiandai field, and Mailang field were 0.90, 0.90, and 0.92, respectively (Fig. 13-C, F and I). Among these, the Mailang field exhibited outstanding performance, with an MAE of 9.55 and an RMSE of 13.02, demonstrating the model's strong predictive capability under conditions of rapidly increasing pest density.

Overall, most experimental fields and stages achieved prediction R^2 values exceeding 0.75. Variations in model performance were primarily influenced by the combined effects of field landscape complexity, pest density, and sample distribution characteristics. The results indicate that the model can reliably simulate pest aggregation and dispersal trends under conditions of high pest density and environmental disturbance, demonstrating strong dynamic adaptability and spatial prediction stability.

4.2 Ecological driving mechanisms of *C. suppressalis* population dynamics

Temperature is a key environmental factor regulating the growth and development of *C. suppressalis*, with a suitable developmental range of 15-26 °C (Shamakhi *et al.* 2018). In this study, DT, integrating air temperature and humidity, showed high importance during the population establishment stage (Fig. 8-B), capturing the combined effect of thermal and moisture conditions on early pest dispersal and accumulation. SM was consistently retained across the expansion and outbreak stages, reflecting its role in influencing pupal development and spatial dispersal patterns (Dai *et al.* 2023). AT, despite being traditionally recognized as a direct driver, was not included due to relatively low spatial heterogeneity within the study area and its indirect influence primarily via developmental rates and generation times (Lu *et al.* 2014).

During the population expansion and outbreak stages, AP and WS emerged as dominant factors (Fig. 8-C and D). AP serves as a regulatory signal for adult activity and migration, with decreases indicating approaching adverse weather and triggering dispersal (Gao *et al.* 2008). Coupled with WS, these factors collectively shape migration routes, distances, and landing sites, driving the dynamic spatial patterns of pest populations (Liu and Ye 2011).

Rice remote sensing physiological indices provide essential indicators of canopy growth and stress responses, supporting spatiotemporal pest prediction. Among these, EVI, reflecting photosynthetic capacity and vegetation health, was consistently the most important during the establishment stage (Fig. 8-B), capturing early canopy anomalies and potential outbreak areas. Additional indices such as BSI, SIPI, and RDVI collectively describe vegetation trait variations and exhibited stage-specific importance (Fig. 8-A), allowing indirect identification of pest-affected areas and severity (Sudha Rani *et al.* 2018).

Overall, by integrating key meteorological, soil, and vegetation indices within a stage-specific predictive workflow, this approach not only provides mechanistic insights into pest–crop interactions but also offers a transferable methodology for forecasting populations of other pests or in different cropping systems. The workflow can accommodate stage-specific ecological drivers, environmental heterogeneity, and remote sensing indicators, making it broadly applicable for precision pest management and dynamic risk assessment.

4.3 Limitations and future directions

Limitations of MODIS data In this study, the prediction model based on MODIS vegetation indices demonstrated relatively good performance at the regional scale. Nevertheless, the inherent limitations of MODIS in terms of spatial resolution should be acknowledged. MODIS 500 m pixels are prone to mixed-pixel effects in highly fragmented Asian rice landscapes, where the spectral signal may simultaneously include target fields, neighboring crops, roads, ditches, and residential areas. This can lead to saturation or smoothing of vegetation indices, potentially obscuring subtle stress signals induced by early pest infestations.

At this level, higher-resolution data sources, such as Sentinel-2 (10 m), UAV-based multispectral, or hyperspectral imagery, offer significant advantages. Specifically, these data can more accurately capture within-field spatial heterogeneity and enable stress monitoring at the field or sub-field level. Additionally, hyperspectral or multispectral information can be used to construct crop stress indices targeting pests and diseases, thereby enhancing the detection of early infestations.

It should be noted that remote sensing data inevitably suffer from temporal gaps due to cloud cover, rainfall, or sensor anomalies. While Savitzky-Golay filtering can effectively smooth vegetation index time series, reduce noise, and preserve phenological trends, it cannot fully compensate for missing observations and may attenuate short-term or abrupt crop stress signals when over-smoothed.

Therefore, future studies could consider integrating long-term MODIS data with high-resolution remote sensing imagery to leverage the advantages of both, achieving a monitoring system with broad spatial coverage and fine field-level resolution. Furthermore, more advanced interpolation and temporal reconstruction methods could be explored to address missing data issues, improving the temporal reliability of the data and the stability of predictive models. These efforts would contribute to establishing an efficient and precise dynamic monitoring and prediction framework for rice pests, providing scientific support for precision agriculture management.

Model limitations The overwintering population base is a critical factor determining the number and timing of the first-generation adult emergence in the following spring (Subedi *et al.* 2023), influencing the early accumulation of pest sources. In 2023, Xiandai field implemented chemical control measures against *C. suppressalis* that significantly reduced the overwintering population base. Because of this human intervention, the actual overwintering population was extremely low, lacking effective pest source accumulation in spring, which caused the actual occurrence time of *C. suppressalis* to lag significantly behind model predictions. The model used in this study primarily relies on vegetation index, meteorological, and environmental variables, but it does not incorporate information on field-level human interventions or pest source background parameters, such as the overwintering population base into account. Consequently,

the model failed to reflect the suppression of pest sources under intensive control conditions, leading to an overestimation of early-season pest abundance at the Xiandai field (Fig. 13-D). This finding highlights the necessity of explicitly incorporating overwintering population dynamics as an important ecological driver in future model refinements.

The multifactor coupling prediction model constructed from rice remote sensing physiological indices, meteorological, and soil factors captures overall population trends but still exhibits noticeable fluctuations in daily-scale predictions, particularly during rapid population-change stages (Fig. 12). Based on the RF algorithm, the model has strong nonlinear fitting capabilities. Nevertheless, during stages of drastic pest population changes, minor perturbations in daily environmental factors input can be amplified by the model, resulting in notable fluctuations in prediction outputs. This instability becomes particularly evident during the population expansion stage in Mailang field (Fig. 13-H). The sensitivity is partly attributed to RF's tendency to overfit a few high-value samples in the training data (Barreñada *et al.* 2024), and partly limited by imbalanced historical data distribution (Gu *et al.* 2022).

In summary, the weaker generalization performance during the expansion stage arises from two main aspects. First, unaccounted ecological factors such as human intervention and overwintering pest-source parameters. Second, model sensitivity to input perturbations during dynamic population phases. Therefore, future research will explore more robust deep-learning architectures (e.g., TabPFN, STGNN, Temporal Fusion Transformer) and develop adaptive anomaly-detection mechanisms. Moreover, continuously accumulating multi-year monitoring data and conducting cross-year validation will help systematically evaluate the model's generalization ability under varying ecological and management scenarios, thereby further enhancing its stability and field-application reliability.

5. Conclusion

To address the limitations of existing pest prediction models in capturing temporal succession and fine-scale population dynamics, this study introduced LOESS smoothing and slope-change analysis to identify stage-specific characteristics of *Chilo suppressalis* population development, and established a stage-based dynamic prediction framework by integrating meteorological, soil, and rice remote sensing physiological indicators. The proposed framework effectively characterizes the temporal evolution and spatial distribution of *C. suppressalis* populations, providing a practical foundation for pest monitoring, early warning, and stage-specific management. Future work will focus on integrating high-resolution remote sensing data, developing advanced deep learning and adaptive anomaly-detection methods, and enhancing model generalizability and operational reliability through long-term monitoring and cross-scenario validation.

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Declaration of competing interest

The authors declare that they have no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used Deepseek to assist in improving the clarity and readability of the language. All content generated was carefully reviewed and revised by the authors as necessary, and the authors take full responsibility for the final content of the article.

Ethical statements

All applicable international, national and institutional guidelines for the care and use of animals were followed.

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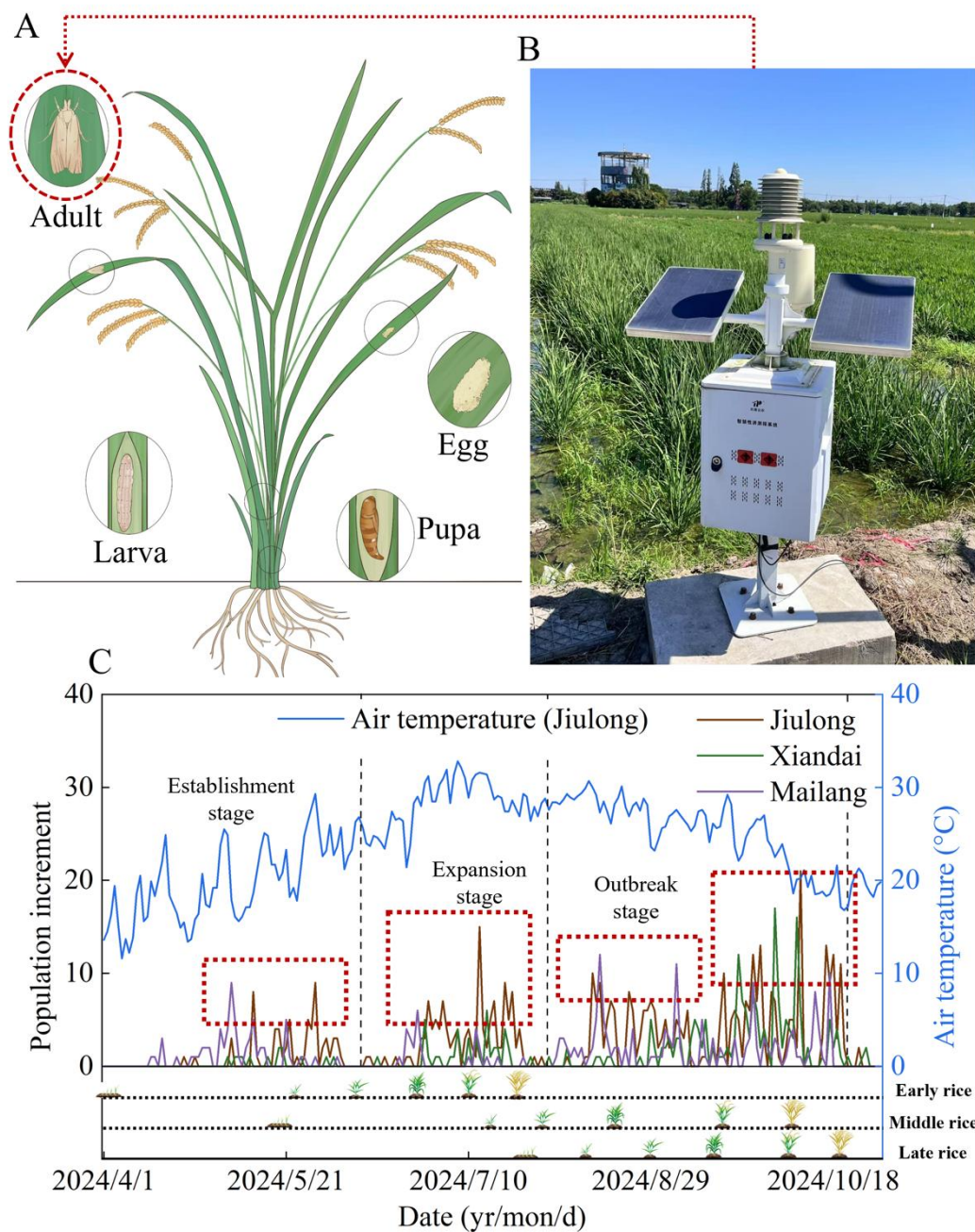


Fig. 2 Morphological characteristics of the four developmental stages of *Chilo suppressalis* (A), IoT-based pest pheromone monitoring system (B), and monitoring data of air temperature and daily increase in *C. suppressalis* population in 2024 (C). Red rectangles highlight the developmental periods of *C. suppressalis* generations 1–4.

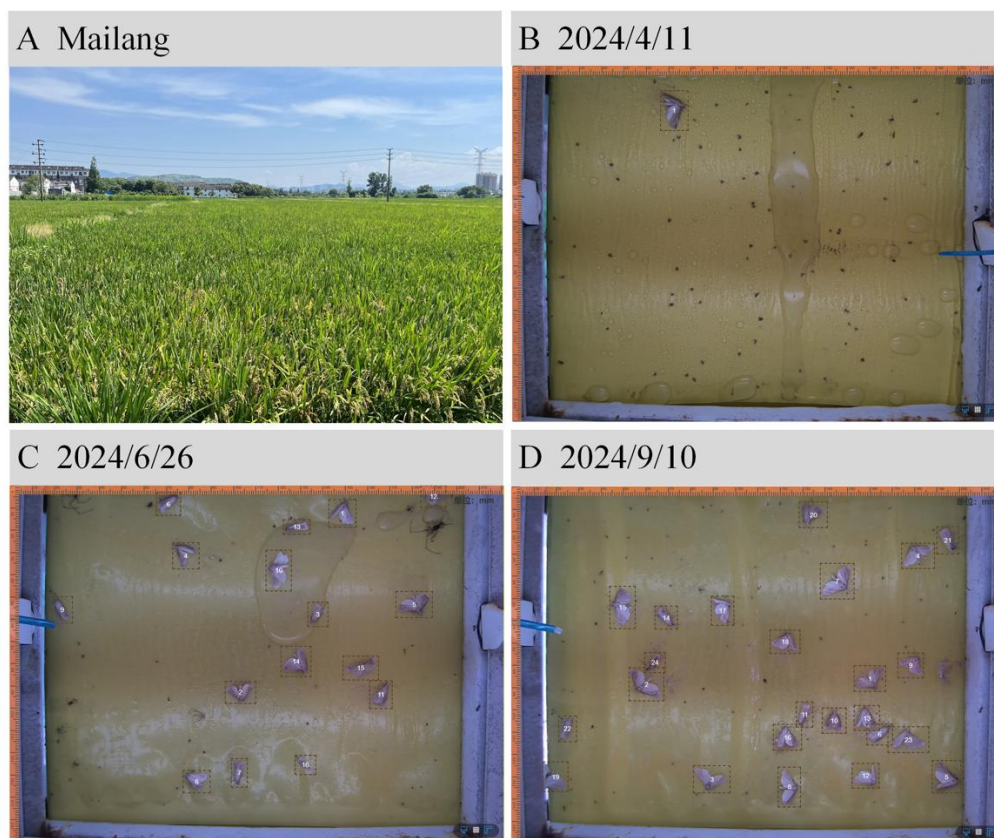


Fig. 3 Mailang experimental field site (A) and automatically recognized images of *Chilo suppressalis* captured by the pheromone monitoring device (B–D).

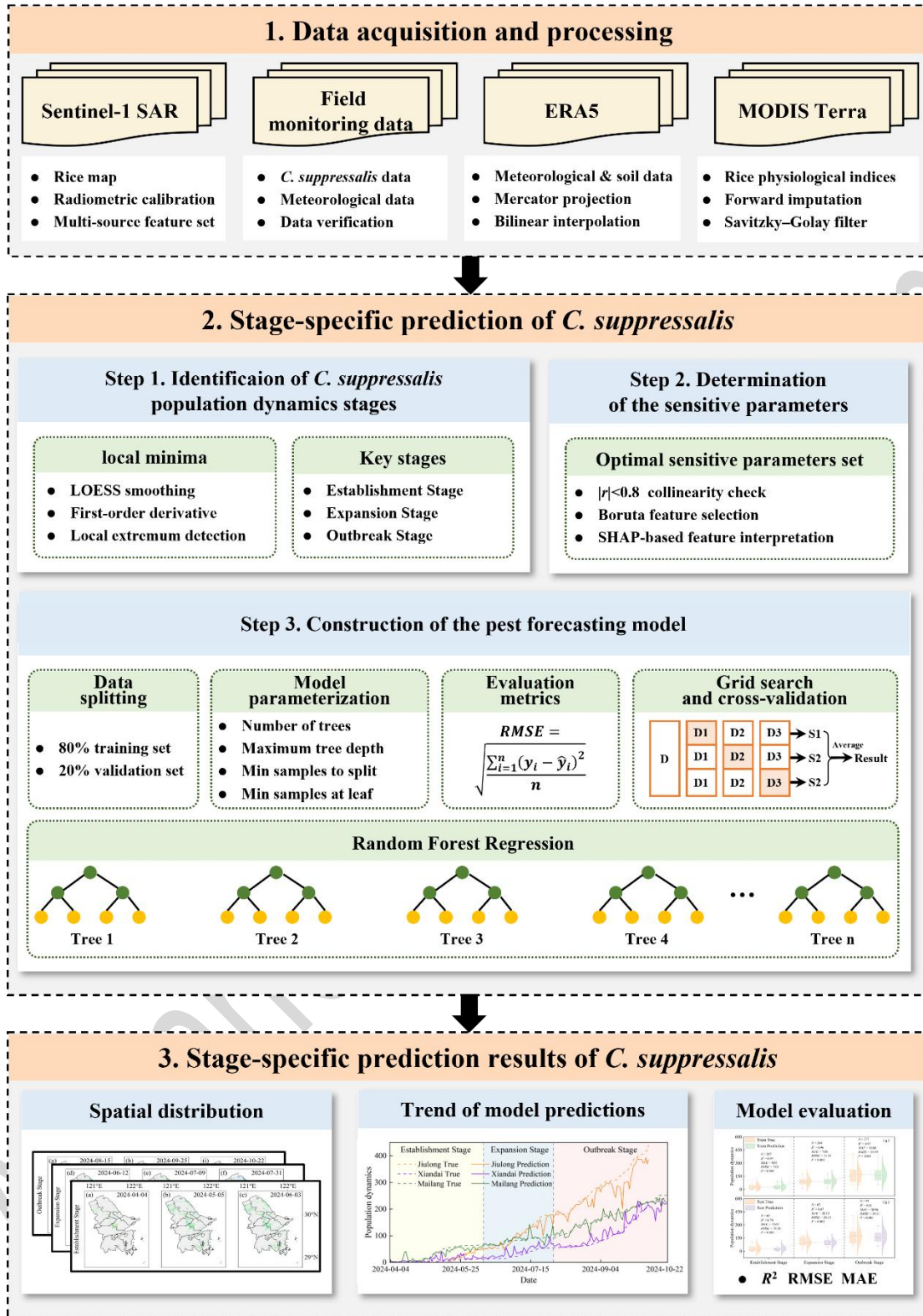


Fig. 5 Methodological framework for the spatiotemporal prediction of *Chilo suppressalis*.

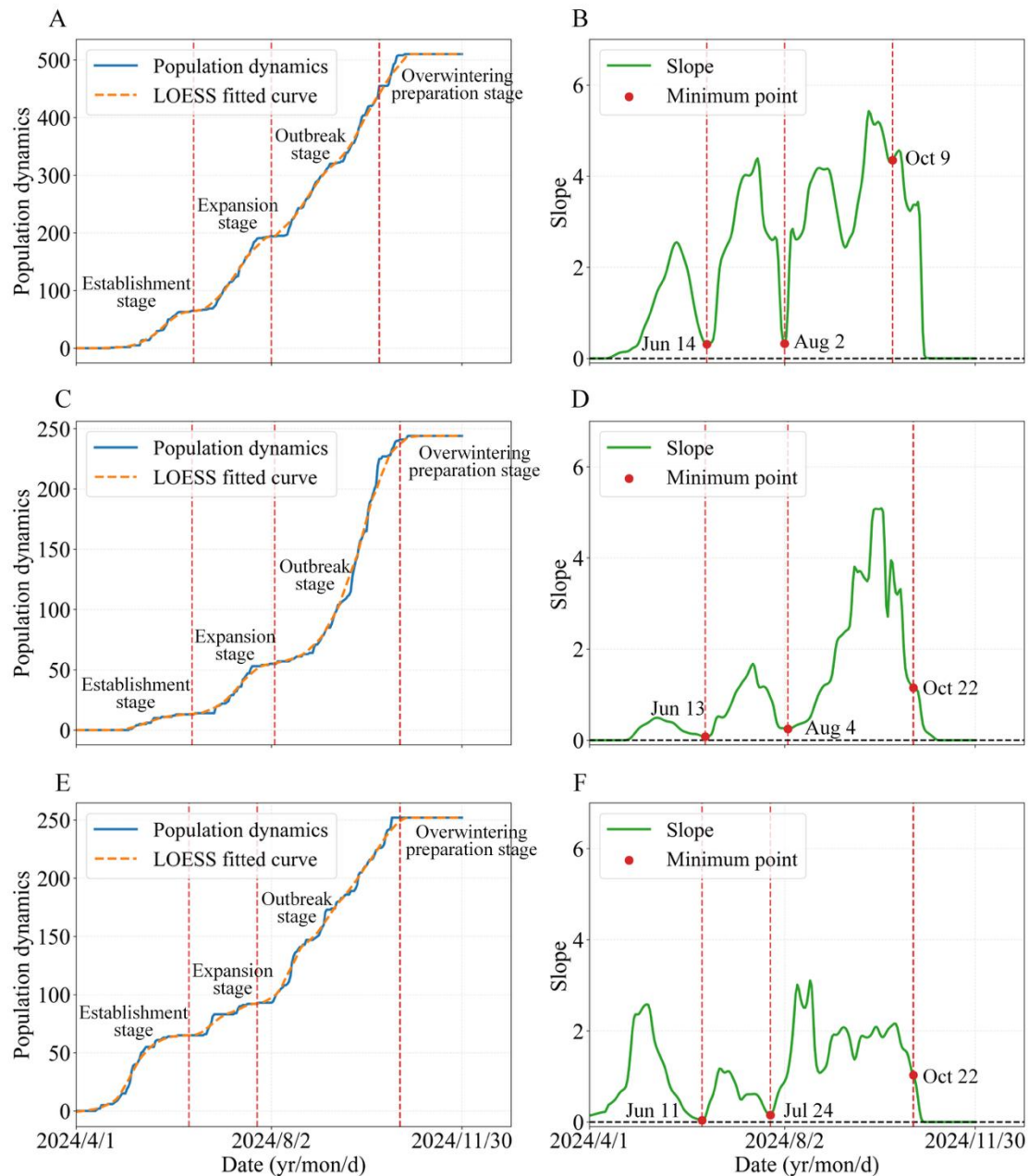


Fig. 6 Occurrence stages of *Chilo suppressalis* based on LOESS-smoothed slope variations. A and B, Jiulong field. C and D, Xiandai field. E and F, Mailang field.

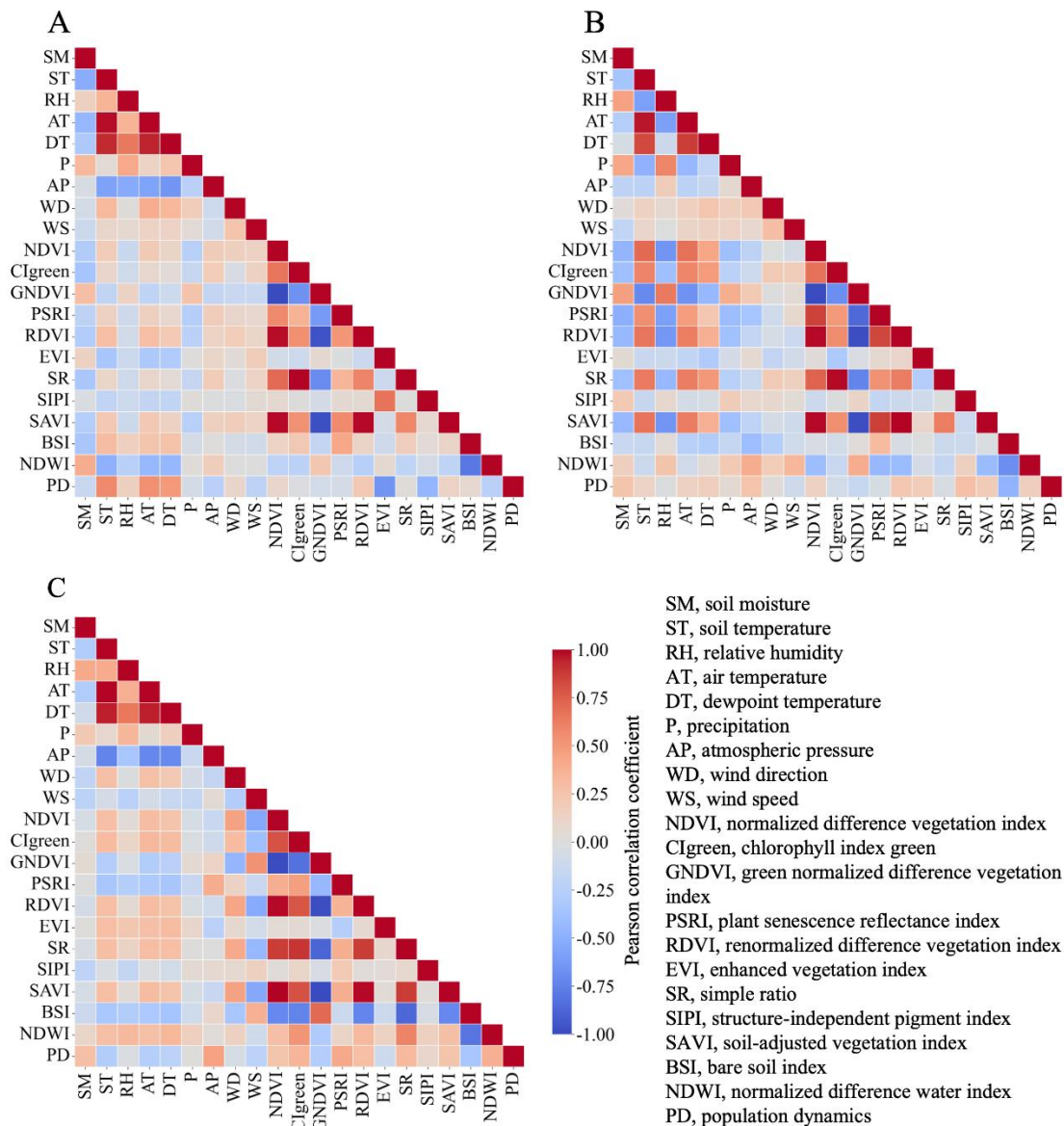


Fig. 7 Stage-specific correlations of *Chilo suppressalis* population dynamics during the establishment (A), expansion (B), and outbreak stages (C).

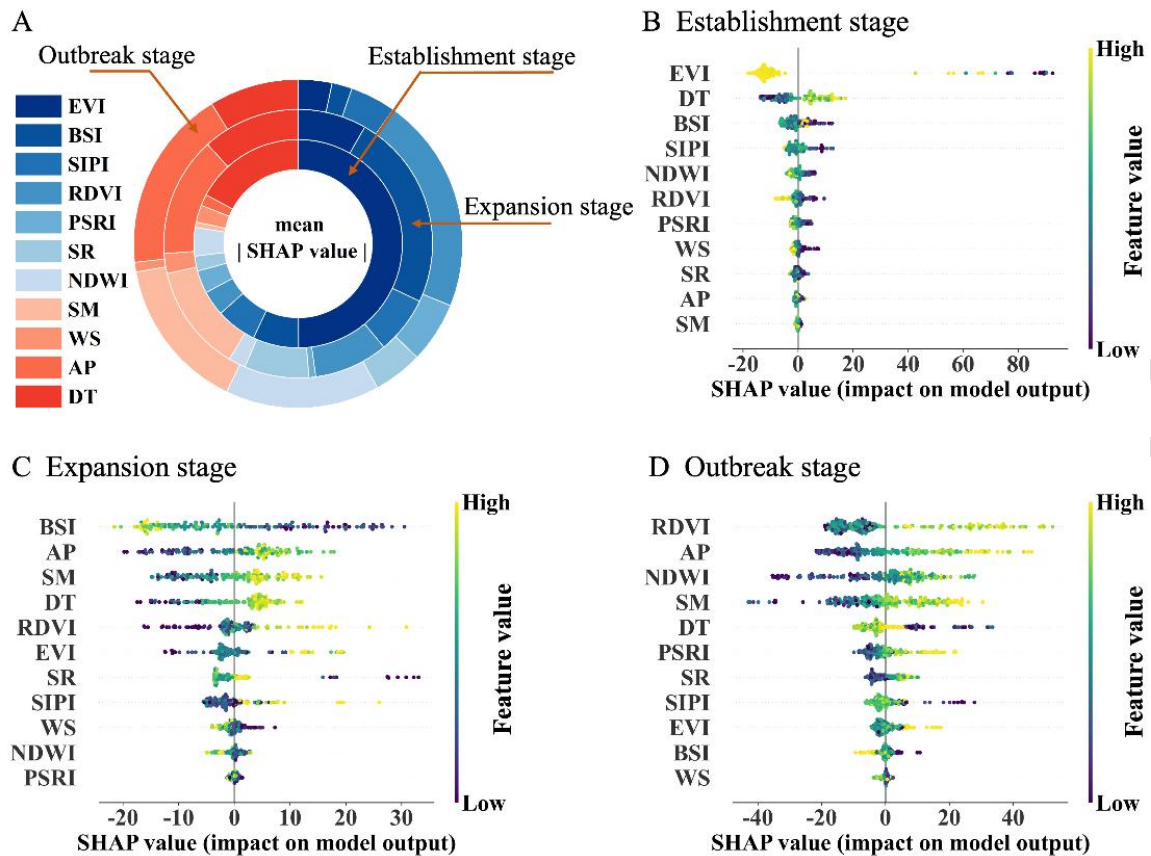


Fig. 8 Visualization of the importance of sensitivity factors. SHAP feature importance rankings across the three stages, with blue indicating remote sensing indices and orange indicating meteorological environmental factors (A), and SHAP value scatter plots for each stage (B–D). EVI, enhanced vegetation index; BSI, bare soil index; SIPI, structure-independent pigment index; RDVI, renormalized difference vegetation index; PSRI, plant senescence reflectance index; SR, simple ratio; NDWI, normalized difference water index; SM, soil moisture; WS, wind speed; AP, atmospheric pressure; DT, dewpoint temperature.

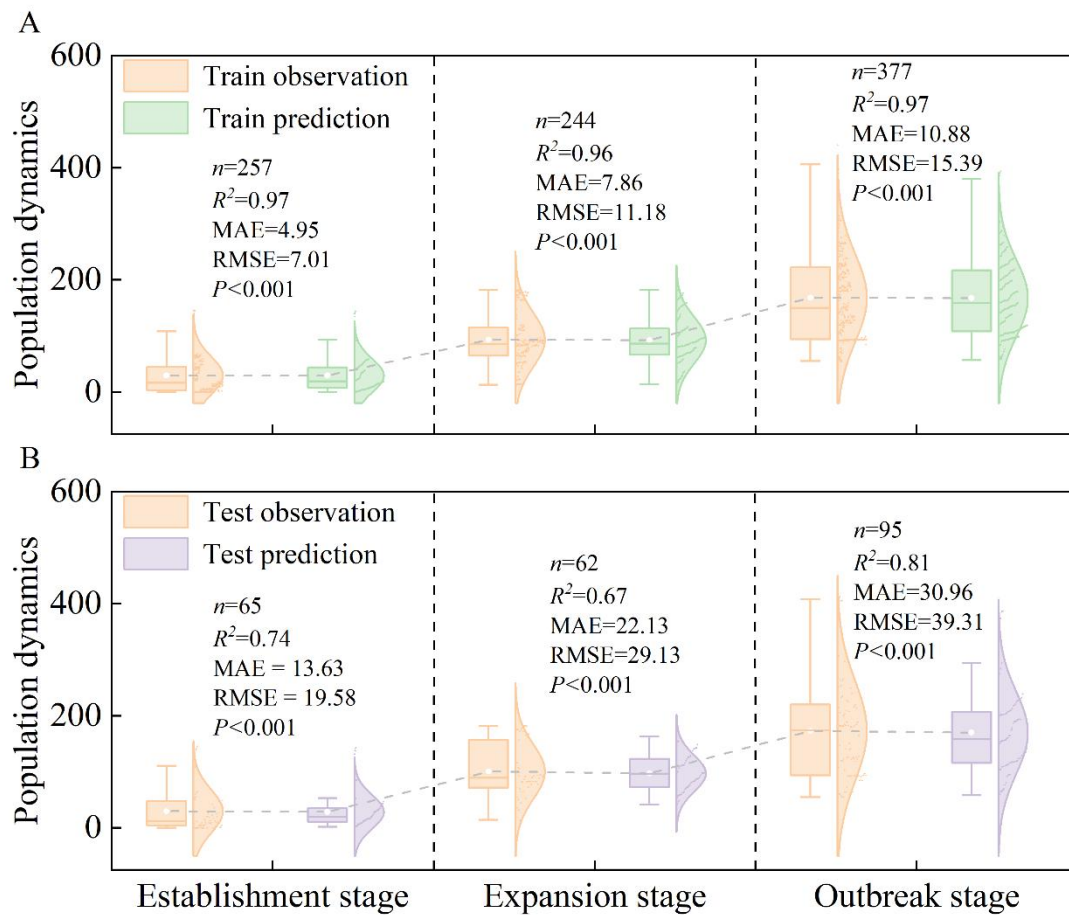


Fig. 9 Box plots of stage-specific model prediction performance for the training set (A) and test set (B).

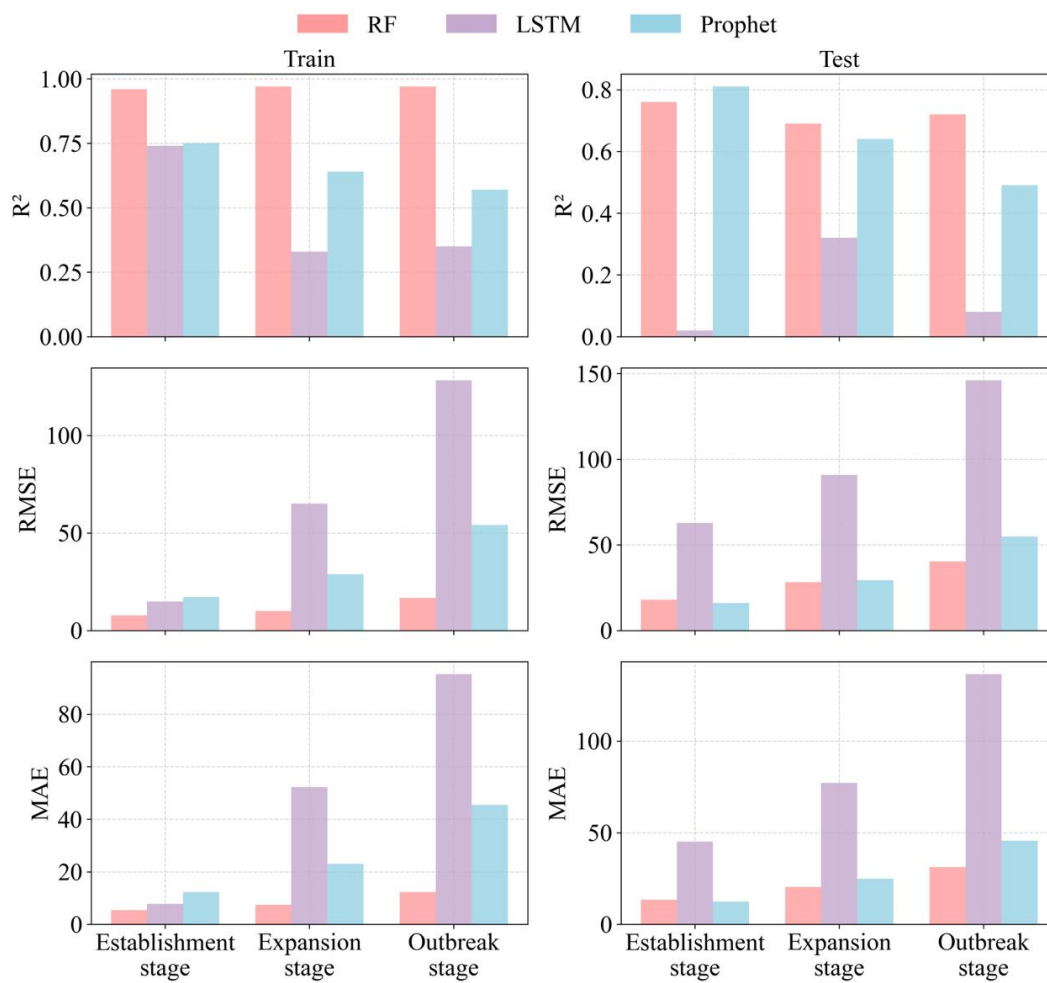


Fig. 10 Comparison of model prediction performance across different stages. RF, random forest; LSTM, long short-term memory.

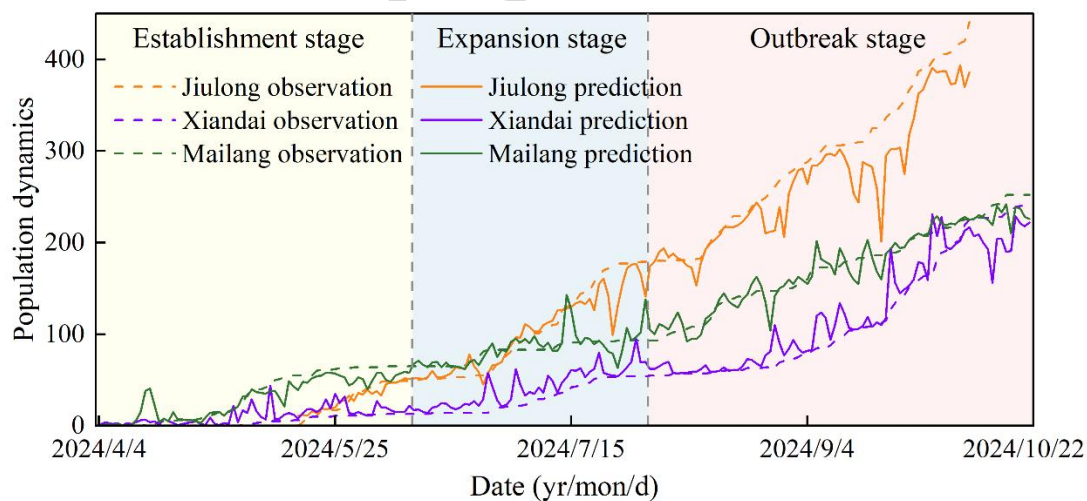


Fig. 12 Temporal dynamics of observed and predicted *Chilo suppressalis* populations.

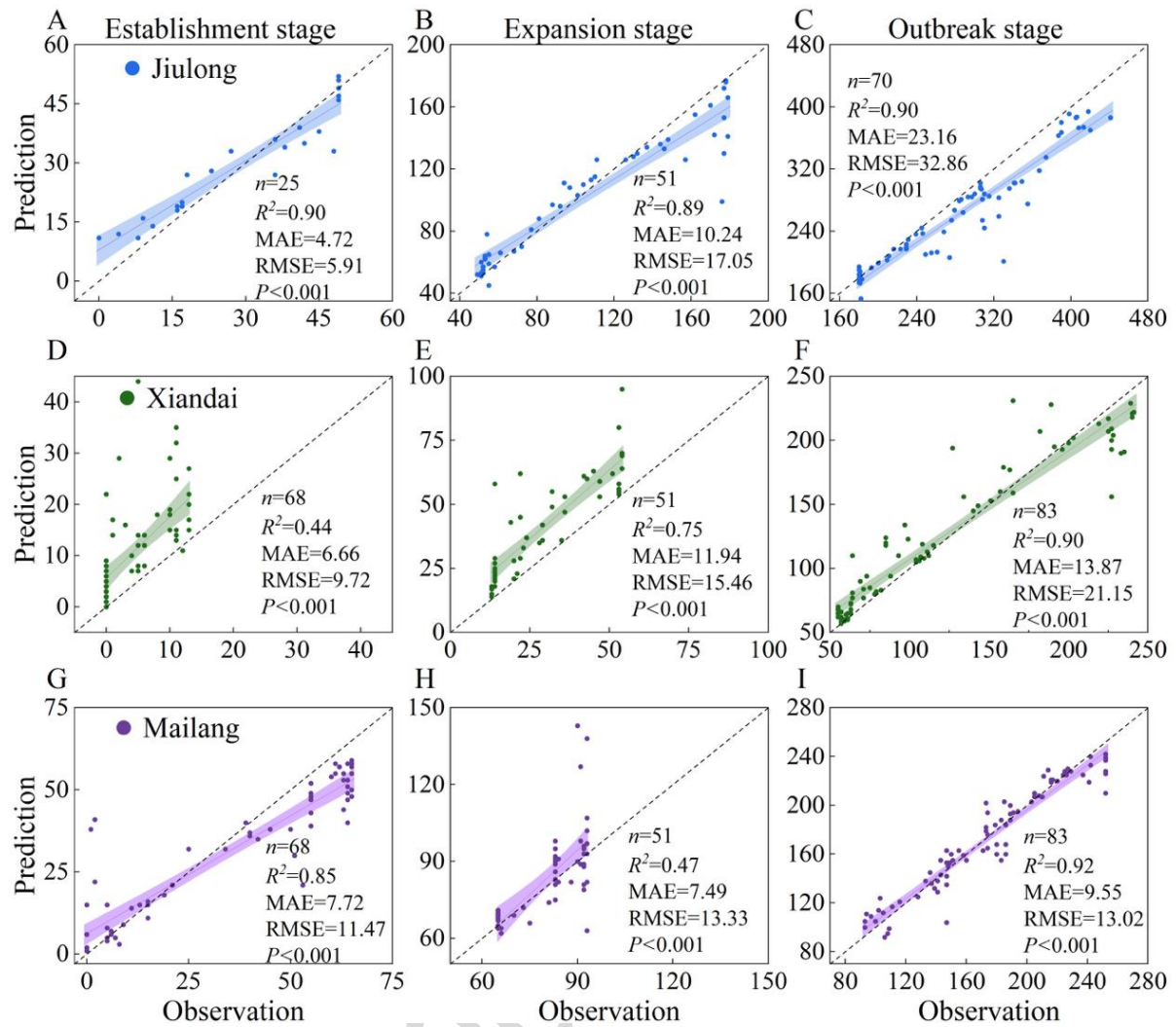


Fig. 13 Stage-specific performance of the *Chilo suppressalis* prediction model across experimental fields, including the Jiulong field (A–C), Xiandai field (D–F), and Mailang field (G–I).

Table 1 MODIS-based rice remote sensing physiological indices and environmental indices

Indice	Formula	Function	Reference
Normalized difference vegetation index (NDVI)	$NDVI = (NIR - Red) / (NIR + Red)$	Monitoring vegetation density and growth status	Rouse <i>et al.</i> (1974)
Chlorophyll index green (CIgreen)	$CIgreen = (NIR / Green) - 1$	Estimation of chlorophyll content	Gitelson <i>et al.</i> (2003)
Plant senescence reflectance index (PSRI)	$PSRI = (Red - Blue) / NIR$	Monitoring vegetation senescence and yellowing	Merzlyak <i>et al.</i> (1999)
Renormalized difference vegetation index (RDVI)	$RDVI = (NIR - Red) / (NIR + Red)^{0.5}$	Vegetation density variation	Roujean and Breon (1995)
Enhanced vegetation index (EVI)	$EVI = 2.5 \times (NIR - Red) / (NIR + 6 \times Red - 7.5 \times Blue + 1)$	Improved NDVI, reduces atmospheric and soil interference	Huete <i>et al.</i> (2002)
Green normalized difference vegetation index (GNDVI)	$GNDVI = (NIR - Green) / (NIR + Green)$	Sensitive to chlorophyll content and Leaf area index	Gitelson <i>et al.</i> (1996)
Simple ratio (SR)	$SR = NIR / Red$	Reflects vegetation cover and photosynthetic activity; higher values indicate healthier vegetation	Jordan (1969)
Structure-independent pigment index (SIPI)	$SIPI = (NIR - Blue) / (NIR - Red)$	Monitoring chlorophyll and canopy structure changes	Penuelas (1995)
Normalized difference water index (NDWI)	$NDWI = (Green - NIR) / (Green + NIR)$	Extraction of water bodies; suppresses vegetation and soil signals	McFeeters (1996)
Soil-adjusted vegetation index (SAVI)	$SAVI = (NIR - Red) / (NIR + Red + L) \times (1 + L)$	Vegetation index adjusted for soil background effects; suitable for low vegetation cover areas	Huete (1988)
Bare soil index (BSI)	$BSI = [(Red + SWIR) - (NIR + Blue)] / [(Red + SWIR) + (NIR + Blue)]$	Enhances bare soil information; used for soil, buildings, and other feature identification	Rikimaru <i>et al.</i> (2002)

Table 2 Parameter settings for the RF model

Parameter name	Description	Description
n_estimators	Number of decision trees	[50, 100, 150, 200]
max_depth	Maximum depth of each tree	[None, 10, 20, 30, 40]
min_samples_split	Minimum number of samples required to split an internal node	[2, 5, 10, 20]
min_samples_leaf	Minimum number of samples required at a leaf node	[1, 2, 4, 6]

Table 3 Screening scheme for stage-specific sensitive parameter datasets of *Chilo Suppressalis*

Stage	Factors removed by $r > 0.8^{1)}$	Factors removed by Boruta ²⁾	Selected factors ³⁾	Final determined factors
Establishment Stage	AT, DT, GNDVI, NDVI, Clgreen, SAVI	SM, RH, AP, WD, WS	ST, RDVI, EVI, SIPI, BSI, P, PSRI, SR, NDWI	RDVI, EVI, SIPI, BSI, PSRI, SR, NDWI, SM, DT, AP, WS
Expansion Stage	ST, AT, GNDVI, PSRI, NDVI, Clgreen, SAVI	RH, P, WD	SM, DT, AP, RDVI, EVI, SR, SIPI, BSI, NDWI, WS	RDVI, EVI, SIPI, BSI, PSRI, SR, NDWI, SM, DT, AP, WS
Outbreak Stage	ST, AT, NDVI, GNDVI, SR, RDVI, SAVI, BSI	P, WD	SM, DT, AP, Clgreen, PSRI, EVI, SIPI, NDWI, RH, WS	RDVI, EVI, SIPI, BSI, PSRI, SR, NDWI, SM, DT, AP, WS

¹⁾ AT, air temperature; DT, dewpoint temperature; GNDVI, green normalized difference vegetation index; NDVI, normalized difference vegetation index; Clgreen, chlorophyll index green; SAVI, soil-adjusted vegetation index; ST, soil temperature; PSRI, plant senescence reflectance index; SR, simple ratio; RDVI, renormalized difference vegetation index; BSI, bare soil index.

²⁾ SM, soil moisture; RH, relative humidity; AP, atmospheric pressure; WD, wind direction; WS, wind speed; P, precipitation.

³⁾ EVI, enhanced vegetation index; SIPI, structure-independent pigment index; NDWI, normalized difference water index.

Table 4 Optimal pyperparameter settings of the RF model for each stage

Stage	max_depth	min_samples_leaf	min_samples_split	n_estimators
Establishment stage	None (14.88)	1	2	200
Expansion stage	10	1	2	50
Outbreak stage	None (16.47)	1	2	200